

Motion & Locking Mechanism Concept (V1)

Design Philosophy

The barrier movement is **mechanically assisted**, while the actuator is used **only for locking and unlocking**. This minimizes power consumption, reduces component count, and improves long-term reliability.

Key Components

- Rigid U-shaped Barrier
 - Dual Pivot Shaft
 - Dual Torsion Springs (Left & Right) – Motion Assist Only
 - Single Steel Rotary/Hook Lock
 - Mini Solenoid (Lock Release)
 - Hall Position Sensor
 - Reinforced Steel Chassis & Base Plate
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Design Principles

- Motor/Solenoid performs **only lock/unlock** operation.
 - Torsion springs assist barrier movement; **they never carry structural loads**.
 - Vehicle impact loads are transferred through:
Barrier → Pivot → Chassis → Base Plate → Ground
 - Lock bears the holding load, not the spring or actuator.
 - Dual torsion springs distribute forces evenly across both pivot shafts.
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Operating Flow

1. Locked State

```
Barrier UP
  |
Steel Lock Engaged
  |
Spring Preloaded
  |
System Secure
```

2. Unlock

App Command
|
Mini Solenoid Pulse
|
Steel Lock Released
|
Barrier Free

3. Opening

Barrier Released
|
Torsion Springs Assist
+
Gravity
|
Barrier Moves DOWN

4. Open State

Barrier DOWN
|
Vehicle Passes / Parks
|
System Waits

5. Closing

App Close Command
|
Return Mechanism Assists
|
Barrier Moves UP
|
Steel Lock Auto Engages
|
System Secure

Design Advantages

- Low power consumption (actuator operates only during lock/unlock)
 - No heavy drive motor or gearbox required
 - Reduced mechanical complexity
 - Lower BOM cost
 - Easy field maintenance
 - High impact resistance
 - Suitable for local fabrication using laser-cut steel, welded chassis, and standard mechanical components
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Research Items

- Final return mechanism (manual assist, cable assist, or compact linear actuator)
- Lock geometry optimization
- Spring stiffness selection
- Over-center motion optimization
- Prototype testing and fatigue validation

Note: The exact return mechanism and spring characteristics will be finalized after prototype testing and CAD validation. The overall architecture is designed to keep structural loads away from the spring and actuator while maintaining a simple, low-cost, and robust mechanical system.